

ActInf Livestream #026.2 ~ "Bayesian Mechanics for Stationary Processes"

Livestream | 1:48:27

hello everyone welcome to the active inference lab

we're here on august 17 2021
in live stream number 26.2

on
the second discussion that we're having
as a group on the paper bayesian mechanics for stationary processes

welcome to the active inference lab
everyone we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this page

this is recorded in an archived live stream so please provide us with feedback so that we can improve on our work

all backgrounds and perspectives are welcome here and we'll be following video etiquette for live streams so we can use the raise hand feature in jitsi this short link has the upcoming live stream discussions

here we are on august 17th in our last discussion on this paper 26.2 next week we'll be heading into a new paper and we still have some openings in the coming weeks so if anyone has a suggestion or an author to contact or wants to discuss their own paper just let us know and we can try to make that happen

today in acting stream number 26.2 the goal is to continue learning and discussing and unpacking this awesome

paper that we've been thinking about for the last few weeks bayesian mechanics for stationary processes and we're really appreciative that the first author is here to join with us and co-learn with us have fun with us and this video is just going to be wherever we want to take it so we have some questions prepared but especially if anyone in the live chat has questions that'd be awesome and we can definitely address them

so we can just start with a quick introduction round and feel free just to say hello or to answer one of these questions like what's something that you liked or remembered about the paper and what's something that you're wondering about or want to have resolved by the end of today so i'm daniel i'm a postdoctoral researcher in california

and something that over the last week made me excited to have this discussion was i was speaking with a physicist friend someone who's working with lasers and oscilloscopes and stuff

and i brought up this paper and they it turned out that they actually were using pid control in their research on a day-to-day basis but actually it was new for them to hear about how it was connected to some of these broader topics but as soon as i

brought out the physical printed paper
then they were just so excited and it
was cool to see how they had all this
practical
experience working with pid
and then they instantly just saw that
there was kind of this crossover point
into some broader questions that also
they were curious about but the
day-to-day practice actually hadn't
brought them to those general questions
so it was a and i learned a lot about
pid control and about actually how it
plays out with the knobs and everything
like that i'll pass to dean
yeah good morning um
yeah i
i came away from the paper again trying
to figure out
how to
create some sort of a coherent narrative
around
the the
physical math and the statistical math
that part for me
really got me going away and thinking a
little bit because i think they they go
in different directions but i do think
that they can work in concerts so
yeah i'd be really interested to see how
that
might play out in terms of
what what our
our guest here um
thinks the directions this might be able
to go i'll pass it to stephen
hello i'm stephen i'm based in toronto
um

i work a lot with
community development and uh sort of
spatial meaning making so inferences of
different kinds
i'm curious
how some of the
ways that the mathematics have been done
on this paper
can
help
broaden some of the questions around
ergodicity and non-ergodicity in uh
that's a bit of a thorn sometimes a
weapon
in the inactivist
active influence dialogue so um yeah any
and that's something i'll be interested
in and um i'm going to pass this over to
dave
so i think dave doesn't have audio but
dave thanks for joining and any
questions in the youtube live chat is
awesome so thanks
to everyone who's joined and passed to
lance
thanks daniel and thanks for inviting me
again
uh sam nance i'm a phd student at
imperial and ucl
and i'm really excited to be here and to
engage in this discussion
i'm really looking forward to
feedback about the paper seeing what's
interesting to to pursue in future work
um so yeah really looking forward to
to share this paper with you and and
talk about it
so maybe one lead-in question before we

jump to the content what stage in your
phd are you

and is this like the meat and potatoes
chapter one or is this um a little
cul-de-sac that you found yourself
having to explore

in the course of another research
project

um yeah great question so i'm right in
the middle of my phd i'm two years in to
still have two years to go

and um

ideally i would like to pursue this work
but so i have two supervisors one is a
mathematician

another one is a neuroscientist

and so really the the goal

is to is to have some kind of
mathematical theory of how adaptive
systems work so i think this is really
the the first chapter

i want to push that further

but then there are also other projects
that take up my time so i'm trying to
make all that work together

awesome

so

for this dot two we have some questions
prepared some topics that we know we to
talk about but anyone who asks a
question in a live chat will definitely
have

time to address that

the paper that we've been talking about
is this bayesian mechanics for
stationary processes paper and in the
dot zero and the dot one there's more
information on the primary aims and

claims so check those out if you haven't already

we're just going to

take one more look at the road map

where we've been where we're going

remember we were talking about markov

blankets and basic mechanics how does

active inference have similarities and

differences from

other ways to model dynamical systems

and let's just go straight to the first

question that we have which is going to

help us

review where we've been and then also as

a jumping off point for some of these

extended questions that we've already

written

so a colleague asked me

and said i think it would be good to

expand on the differences of the

paradigms in section 3. and those are

section b c and d

related to a posteriori estimation after

the fact

predictive processing and variational

base so perhaps lance first and then

anyone else who wants to give a take or

ask a follow-up question

what are the similarities and the

differences here and why were they

structured in this order is this the

chronological order is this most

specific to least specific or what is

the ordering and the meaning here

right um

yeah so so the

it goes from the the simplest to the

most sophisticated

um so

uh adversary estimation is a special
case of predictive processing which is
specific

special case of variational base

and so the idea

is

if you

if you receive some piece of data so for
example if you see an image

um with your eyes you're gonna try to
infer what's going on

outside in the environment and so if you
do a posterior estimation

what you're trying to guess is what is
the most likely thing

um like what what is the most likely
thing that's going on in the environment

so it could be if you if you see an
image of a chair you you would say oh
it's a chair

but you wouldn't encode any any kind of
uncertainty oh it might be like an
armchair for example or something a bit
different so in a posterior estimation

you try to guess what's going on outside
but there's no uncertainty

then predictive processing you encode
the uncertainty so mathematically
that's called the variance or the ov
precision

and then variation of bayes it's uh it's
more sophisticated so you actually
encode

uh an arbitrary distribution

uh on external states um so it could be
oh it's very likely that it's an
armchair

and it might also be very likely that
it's a cat so that this example doesn't
make sense but you
you could imagine that there's two
things that are very different that
could be equally likely so for example
if you see an illusion it could be
something but also something very
different so you would imagine these two
things of having
a high probability of being there and
then other things are having low
probability
so this is really um
the the most general thing uh and also
as a side comment so rational base um
like a lot of machine learning
algorithms you can see them as special
cases of variational base so i think
this is really the most general way to
think about inference
and and this is really at the heart of
the free energy principle
and i think this is why people have been
describing the brain as performing
variation of base
because it's really the most general way
to think about this kind of problem of
inference
awesome well
before i give a comment on that what
made you say that it was at the heart of
the free energy principle
right
so the free energy principle
says that the brain is doing a
variational base
and and then all this work i mean this

paper in particular but like

um

the last decade people have been trying

to justify

why would the free energy principle

apply actually

um and so and so this is part of uh this

paper is part of a line of work where we

try to examine physical systems or like

adaptive systems and see whether we can

describe them as performing variational

base

great so first stephen with a question

and then anyone else with a question

thanks that's really helpful that

explanation and i was wondering

is it a case then that you can go only

so far with say predictive processing

um and then to get to variational bays

you could and to bring in that

uncertainty you kind of need a free

energy

principle solution to be able to bring

in these

more unknown models and

to go beyond variational bays at some

point that's when you'd have to bring in

the active inference

to

it would that be a kind of the way you

would think about it

yeah yeah we i would think about it like

this um

i would say uh like loosely speaking the

term predictive processing

i think is also meant uh to refer to

variational base but at least a

variational base that's like a proper

mathematical definition of what it is
and then predictive processing it's more
like oh we're predicting something so
it's not that clear how how
i mean there's no clear definition of
predictive processing uh but in this
paper predictive processing is meant as
predictive coding
as in like the papers by ryo and ballard
where they looked at the visual system
and then they looked at
neurons in the cortex as
forming predictions and
suppressing prediction errors
um so yeah um
if you if you only do predictive
processing in in this sense i i think
you're you're limited
and variational base and active
inference is really like the the next
generation because you can accommodate
all sorts of models
and um and yeah i don't i don't think it
i mean i i cannot see at least uh how
you would go beyond that
thank you so
dean with a raised hand and then anyone
else
no that's i was curious so
it seems like the math is able to
help us understand the difference
between something that's really highly
defined
so that kind of a posterior posteriori
and then the sort of the fuzzy
where the very old variational base part
sort of gives us something
a little more a little less defined in

terms of the edges
and so i was just curious
does the brain kind of go back and forth
to to some degree mathematically
speaking between that which is really
clear and really
has a hard edge mathematically and and
those things which statistically
seem to
distribute within the within the
parameter set
um
yeah i mean i think
all these three um like a posterior
estimation as well as variational base
there's uh displausible explanations for
how the brain would implement them okay
and so so the idea
the idea for um
variational base or predictive
processing is that so the brain would be
encoding some kind of uncertainty
and so in the literature
people say that the the mean firing rate
of a neural population
would encode the posterior estimates so
what you think is actually out there and
then the variance of the firing rates
would encode your uncertainty
so i think
so i think there's possible plausible
theories for how the brain would
implement all of these things
and then i think the that's the nice
thing at least with active inference is
that we try to develop models
that are consistent with our
understanding of the brain and as our

understanding of the brain evolves we

kind of like refine these models

um so so i think

we're definitely working towards uh

models

that that do explain at least i mean so

far it's very coarse grain and very

simplistic but we're working towards

models

that could be implemented in the brain

so i think like these um

very very abstract very general form of

variational base could be implemented in

the

brain one quick follow-up is you

mentioned how the mean firing rate

corresponds to this

first

a posteriority estimation which is a

mean estimation and then there's the

variance in the firing rate so what

measurements or what does it look like

to implement variational bays in the

brain like what measurements are we

making what kinds of analyses would we

make on those just like mean and

variance

um

well so so the idea i mean the very

simple idea is that you would get some

sensory stimulus so

you you get some some impression on your

retina

uh some neurons in your retina with fire

uh into the into the visual cortex and

then um

the the the premise is that your brain

has

embodies them
a generative model of its environment so
so it's able
it has learned how to recognize uh how
to how to infer things in the
environment from its sensations
and so the
the stimulus that would be that would go
into into v1 and all these other parts
of visual cortex
and then the neurons in in the visual
cortex would uh
align their means firing rates with
respect to oh it's an armchair or it's
something different
um so maybe there's a neural population
that encodes armchair and there's a
neural population that encodes something
else
there's also a lot of studies for
specific neurons like one neuron
encoding one particular stimulus
so these um there's these guys at mit
that show that there's actually a bill
clinton neuron uh that fires whenever
you see bill clinton so you can be very
specific
and um and yeah but if it's uncertain
that it's bill clinton your
your bill clinton neuron would like fire
yeah i guess would have a lot of
variance in its firing rate
okay awesome so welcome
blue with a raised hand go for it and
then um we'll continue and also steven
so go ahead blue
um i am blue knight and i'm an
independent research consultant from new

mexico and i have a question on your paper and it's kind of something that i've been questioning a lot lately with regards to the free energy principle and the requirement for ergodicity um and my question is um in your paper you said that the um you know this the outline even in outline it says you know you you partition the world with the markov blanket and then you equip the partition with stochastic dynamics and so i'm wondering how these stochastic dynamics relate to organicity if at all and if that fulfills the requirement for um ergonomy as necessary by the fbp right yeah that that's a great question um yeah i think i think stephen also also wanted to talk about that so it's great that you bring this up um so yeah so so the idea of these stochastic dynamics is that we don't assume that dynamics are deterministic and it makes sense because um you can describe planets as having deterministic dynamics but when you talk about biological systems things are a lot more messy and if you're not modeling them at the molecular level which you which you never are things are stochastic so then the the assumption on our system is is that it's stationary

and this is really what it means to to
be adaptive so you're preserving some
kind of steady state
but we never assume ergodicity
so there's actually um a deep um
so stationary and ergodicity are not the
same so we assume
stationarity but we don't assume
ergodicity and so if daniel you could go
to the appendix b
there's um there's a figure i think
which would be the best way to
to illustrate a bit below
yeah perfect
so
um
here in this appendix we
we illustrate uh some dynamics that are
ergodic and some dynamics that are not
ergodic
so
if you if you look at the
uh top left top and bottom left panels
these dynamics are ergodic and and so
the
i guess the intuitive definition of
ergodicity
is that
here you have the the steady state in
blue so so regions that are very blue
are regions of high probability mass
under the state state of regions where
you
your dynamic is likely to be in and
regions are less blue
you're less likely to be in
now ergodicity means that if you start
your dynamic anywhere

and you and you let it run for a
sufficient amount of time so like you
observe your biological system for a
very long time
um your dynamic would basically sample
the whole distribution it would go
i mean if you if you give it enough time
it would go really far away
and then it will come back and it will
do all sorts of things as you can see in
the bottom left and bottom right
um so this is ergodic now in the
in the in the bottom oh no in the top
left and and bottom left it's it's
ergodic and then the
bottom right it's not ergodic
and so
um so as you can see if you start the
dynamic on the bottom right anywhere
it will go around an orbit they won't
sample the whole distribution so it's
still confined to like a
a small region
and and so this this uh so the the paper
still applies for these kind of dynamics
um
see i don't i don't know does that help
or do you
or was that
not very clear it's so such a follow-up
the paper does apply in
situations where it's not
that's what you said right right we yeah
we don't need aerodynamicity well so even
within like a steady state
um like so biological systems you know
operate
within homeostasis but it's a range

right so so it's not
like a constant it's not like you know
purely we're not at 37 degrees all the
time right like so there's some
fluctuation all the time within that
steady
state and and so i just want to kind of
how you rectify those concepts like is
it like a restricting of the the state
space or or how does that work
um
so so the this idea of homeostasis
that's that's really the the steady
state assumption
so for example um
in the state space maybe you would have
one one uh one coordinate that describes
the temperature of your
of your biological system
and if it's a human um
then you will have regions of very high
probability so here regions in blue
around 37 degrees and then regions that
are less blue
uh like a bit further away
uh and so the dynamic
i mean your your system can evolve in in
all of these regions just like on
average you will be more often around 37
degrees just because that's how we are
and then but sometimes it will be higher
and sometimes it will be lower and and
so this is all allowed
um and so i guess the this ergodicity
assumption so then if you assumed
that the biological system was ergodic
then this would imply that if you if you
leave the biological system

like run for a sufficiently long time
like you observe it for a very long time
it will basically um
it will sample your whole distribution
of temperatures
um
so so maybe
um
so so
maybe like the the temperature that
um a human body is allowed to have are
between like 36 and 39 let's just say
for for simplicity and so ergodicity
would say
um yeah if i if i observed my biological
system for a sufficiently long time
i would see my biological system at all
these temperatures
and i think maybe
maybe that would happen so maybe our uh
maybe temperature is like uh
we're ergotic with respect to
temperature but maybe there's other
things in which we aren't ergotic
and um
and i think the the previous papers on
the free energy principle they
for simplicity they always took like um
they always assumed some kind of dynamic
and it turned out to be ergotic
and so in here we we're not making that
assumption
so
i'm sorry i'm going to just follow with
one more question daniel is that okay
yeah um
so so
you know it's great right if you're

talking about one human system right
like my mark on blanket the partitions
like internal and external states and so
i have my one human system or even
multiple human systems like we will
operate in the same kind of steady state
based range but but if we're talking
really about um you know collections and
how kind of
collective dynamics and overlapping
markup blankets
do you think it's mathematically
possible to extend the state space like
i can operate you know from 36 to 39
that's an okay temperature for me but
like a reptile has a very different kind
of temperature range but clearly like we
have some interaction i have interaction
with the reptile and so there's some
some emergent dynamic between me and
some reptile future right right like so
can we define a state space like would
it be mathematically like
acceptable or is it like offensive
mathematically like
impossible to
define a state space where i'm allowed
to sample all of these temperature
ranges with my markup blanket but some
other feature in some kind of collective
dynamics is not like they have a
different range like is that still
organicity because it's not clearly i'm
not sampling the entire state space i'm
sampling my state space but like is that
confounding in some way i mean it is in
my brain that's why i'm asking yeah that
that's a really good question um so so

this is all fine so so we can do
everything you just said
and so when we um so actually when
when we say oh we so we we always sample
a distribution
we're not i mean and then the
distribution is on the safe space so
we're sampling the state space
but like if your distribution is
confined to a region of state space so
let's say
as a human um i cannot possibly have any
bodily temperatures below 36 degrees
then your then you would sample only
things are above 36 degrees
um so i think you would have your own uh
steady state your own stationary
distribution with temperatures around 37
degrees and then if there's a if there's
a snake also in the in this environment
that you're modeling then we'll have a
another very different steady state
and so
i think it really
uh your question really speaks to the
possible extensions of these of this
work
because in this paper we just say oh
there's like an internal state space
which would be what's inside
and then there's an external state space
which would be what's outside
but then um
really what what people would need to do
now is to say oh but my
actually there's a lot of very
interesting things in my environment
there may be snakes there may be

other kinds of animals and they all have their own markov blanket and they all have their own steady state so then what you can do is to partition your external state space into all of these things that might be around you and then instead of having some kind of um performing some kind of simple inference as we have in this paper you would have some more structure inference a more structured inference because part of your inference would be about what's just in front of you part of your inference would be about your dog that's right over there and you would have a lot of by carving your external state space into different things with their own steady state you would actually have a an inference that's much more structured because you would infer all these things all these different things that are in your environment so this can all be accommodated in this paper but then you would definitely need another paper to like make that explicit and and say some really interesting things because yeah this is really um the way to go but the the mathematics doesn't need doesn't need extension it's just like um yeah it would just be like partition the state space and then yeah we can have

all that that you spoke about
thank you
great and also one piece so then
stephen's question is it's it's not as
if something being ergodic or not is
equivalent to it being easy or simple or
tractable like looking at these
distributions um or trajectories here
the one that you characterized as
non-ergotic not sampling um basically
across the state space the bottom right
this one
is the one that in the caption it says
purely conservative dynamics lower right
panel are reminiscent of the
trajectories of massive bodies whose
random fluctuations are negligible
so it's actually not like it's um maybe
this will speak to steven's questions
about what is a thorn and and what's a
rose
because this is a non-ergotic system
that's
massively predictable
you can do it with a you know a gear
so
maybe that relates to
some of these criticisms about well
we're only working with a limited set of
ergotic processes it's like okay but
that's the squiggly one that's dancing
everywhere the stochastic process that
we care about whereas including
non-ergotic processes some of these
non-organic processes might just be
going around in an oval so it won't be a
challenge for the math at all something
that works on the stochastic squiggle is

going to have actually an easier time on
this oval one so
stephen and then anyone else with a
question

yeah thanks um

yeah i suppose actually tying into what
you just said there daniel the
that there's this question with um
how much we

need to be able to sample over big state
spaces so it might well be that i don't
know the chemicals of a cell there
with molecules vibrating
millions of times a second and
there may be a you know a quick
ongoing way to sort of get a feel for
the ergodic state space

now if you watch a
cricket ball come towards you
um

you know you don't get that chance to do
that so but you might want to sort of
maintain a trajec some sort of
trajectory

um that can still be um

tractable i suppose so

i think this is this is actually quite
quite useful

um i suppose the question then comes in
is

how what's the sort of in-betweens you
know where does

um approximation science come in more
and more

to allow us to sort of

work with not knowing

um and work with something like um these
solenoidal flows

um
in a more fuzzy way um
because i am assuming that if you've got
less chances to sample your state space
solenoidal might be
more useful
to get towards something tractable or
maybe i'm wrong but i wonder what your
thoughts are on that
right um so actually solenoidal um
i mean so
in in this figure here uh on the bottom
left you have like
no solenoidal flow and then the bottom
right there's just so the node or flow
um so i would say
all of these two like very special cases
uh they don't look like biological
systems at all
and they're also the the simplest ones
to to understand mathematically and to
study mathematically
and actually when it gets really
interesting and when it looks a lot more
like a biological system is on the top
left
and that's harder to
to assess mathematically um
and so as you can see it's quite i mean
i think the dynamic on the top left is
quite nice because on the one hand you
have dissipation like you have some
squiggly lines you have randomness
on the other hand thanks to the
solenoidal flow you have some kind of
exploration
and also some kind of like circular
patterns so for example in um

and this is actually how how you correct
there is what's called a non-equilibrium
steady state but without without going
into the details
um as humans as and as animals we have a
lot of circular patterns they are not
purely deterministic so for example we
all have a circadian rhythm
um and we all have all sorts of other um
yeah circular patterns and i think
yeah to account for that uh you really
need to be to be on the top left
um so i think the

[Music]

i mean one of the next steps to this
research is
right right now we just assumed that the
dynamic is stationary and so all this
applies to like a humongous
range of dynamics and a lot of them
don't look like biological systems at
all

and i think the next step would be to
to restrict ourselves to dynamics that
look a lot more like that like
biological systems

and then see what what else can we say
and and i bet there's some dynamics that
look a lot like very intelligent systems
and some that also look like maybe
lifelike systems but they are not so
intelligent so the question is what kind
of parameters what kind of dynamics do
you need

um to characterize something that
explores its environment uh something
that makes decisions and all sorts of
things

um

yeah

and and that's going to be a challenge

and also

i mean just speaking about biological systems i think biological systems have a lot of solenoidal flow and they do have dissipation but they don't have so much dissipation

so you would have something like the top left but with like huge

you have huge like a lot of expiration like you would go a lot further than that in your state space

and then you would have a bit of i mean you would have a lot of squiggly lines but

it would be yeah there would be a lot more like exploration than there is dissipation

yes stephen go forward

yeah and and

you have both can combine can't so gradient

sin and solenoidal flow can kind of be playing at the same time i'm wondering if someone was to bring attention to a goal or a type of attention maybe more pragmatic attention or an epistemic attention or a risk adverse attention

whether

it might be that rather than trying to deal with the whole state space maybe that that general

process carries on but in it tries to introduce a solenoidal

you know um type of flow which in some

ways you could imagine being more tractable to a biological system if it's like bioelectricity

um there's you know the types of dynamics which may be are more tractable around a certain trajectory um than some

um so you've the these the these dynamics could combine

um

in different regimes of attention

yeah

yeah definitely um

and i think but i think to to properly account for attention we we need to go beyond these continuous state spaces and do what people um

um and basically extend to these standard models of active inference where in in addition to have some continuous state space maybe like for movements and this kind of thing you also have a discrete state space and then you would have a dynamic evolving in your discrete state space and maybe like

some state would be oh i'm attending over there and some other state would be unattending over there so i think attention is really something

um that can be modeled with a discrete state space

um

and then it would

i don't think it would be so hard and i mean at least seems doable to extend all the math here to like models

uh yeah that have some like continuous

state space maybe for movement or

or

or other kinds of things like maybe like

blood glucose concentration is also

something that evolves on a continuous

scale there's also a lot of a lot of

things in our mind that are discrete so

is this red yes or no

is it raining yes or no

and to account for all that we need uh

we need a discrete state space as well

awesome thank you lance welcome scott i

see your hand raised so go for the

question and then anyone else with a

raised hand

thank you i came in a little late um my

apologies is fascinating the solenoidal

flow when that gets set up

is it possible that he had multiple

state spaces and

might because of the solenoidal flow

might it have a periodicity in each of

the state spaces so that you might be

able to use fourier transforms to unpack

multiple state spaces in a biological

system

so could you use

having that periodicity does that allow

for a fourier type analysis to start to

unpack state spaces potentially when you

have multiple state spaces and you're

trying to distinguish them

yeah that's a that's a really good point

so i know um

i know a paper actually that that

analyzes the solenoidal flow in terms of

the fourier transform

um i haven't really really thought about

it i think i mean if if you get rid of
the dissipative part so if you get rid
of the squiggly lines and you're just in
the bottom right

then you can definitely use fourier
transforms

but then also the dynamic on the bottom
right is not is not representative of
biological systems

um

so then what you can do when you're on
the top left which is really the
interesting case

is analyze the time irreversible part so
the solenoid will flow with the fourier
transform

and then so you so you do this hem holes
decomposition that we're talking about
in the appendix

uh you decompose your dynamic into
yeah so it's the equation a^2

um

yeah yeah perfect so you decompose your
uh your drift b so so b is the direction
in which your dynamic is moving u_m on
average into reversible and irreversible
parts the irreversible part is the
solenoid overflow which you can analyze
using fourier transform

but then you need to like figure out how
to accommodate the reversible part with
it

so i can share i can share the paper

um

if you're interested but yeah it seems
like like they're they're able to to
like do do some things with that
technique uh but it's also pretty pretty

involved to be honest

a question on the time reversible and
irreversible

what does that mean and in a world where
it seems like we're moving forward
through time only what exactly does that
mean that there's something that's
invariant under time reversible
versus something that changes sign under
time reversal like if we can't
get the time machine or put the process
in a time machine why does it matter
that it's time reversible or how does
that fall out of the equations
right um

well yeah so so

all this is really to emphasize that
we're not assuming the dynamic to be
time reversible

and and as you say everything in the
world around us i mean most things are
time irreversible so
the

uh if you go back to the to the figure
just below so the movement of planets is
really something that's irreversible i
mean the the earth turns around the sun
in one direction and if you are to
rewind time

it will turn around the sun in the other
direction so that

so in this case you can really tell the
difference between

time as time goes forward and as time
goes backward and that's really what's
meant by time irreversible

um but now if you look like a other cup
of coffee

and the coffee is still
um the the
molecules the
coffee molecules or whatever molecules
you have in that cup they're still
bouncing with each other
and that's that's something that's time
reversible because if you look at the
cup of coffee um if you if i give you a
movie of the cup of coffee sitting still
you can watch the movie uh as time goes
forward or backward and uh and you
wouldn't be able to tell the difference
so that's something that's reversible
and so as you can see from this example
actually things that are reversible
aren't interesting at all
from at least a biological perspective
and there's a there's so much literature
about
um yeah about one of the
uh i mean one of the fundamental things
of being alive
is being time irreversible
it's about like eating and producing
waste um it's about yeah consuming
energy
all that kind of thing all that kind of
stuff um produces entropy and is time
irreversible
um but then time so time irreversibility
it's something that's necessary to be
alive
but it's not it's not sufficient
um so for example if i give you
um a convection cell
though that would be time irreversible
but it won't it won't be alive

so any any kind of cycle that's more or less periodic um so maybe if you if you see like a cyclone
a cyclone is time irreversible because it's like
spinning one way and not the other so if you run out of time it would be spinning the other way
so it's different so it's time irreversible but that's not alive
um and so
there's many
many things like that time irreversibility um
yeah creation of entropy
there's a few things like that that characterize biological systems
and i think as we as we go on with this work we'll we'll try to like specialize to only those dynamics that satisfy all these requirements and eventually things like oh we need systems that are able to reproduce for example like a marker blanket that is able to divide into two markup blankets for example when a when a cell decomposes into into two cells so
so this is really the beginning and and yeah in the next years i think this community is really going to be able to to like address all these challenges and like
um
all these characterizations of living systems to to gradually evolve towards models that are

yeah more realistic i would say

awesome

so

i'll have more to say on it but scott go

for it

so a couple of things there's a book
called life's ratchet which is kind of
interesting just on that um the idea of

that irreversibility and that value of

that the idea is that life kind of

ratchets in these forms that work with

the externality i haven't revisited that

since i've been exposed to active

inference but it might be interesting to

take another look at life's ratchet also

one of the things i

like that discussion of living forms and

what's necessary uh prerequisites to

living forms one of the working

definitions i'll share that i'm using

now

and i think let me tell you why i'm

using it is um that life is auto

catalytic

and entropy secreting

now my kids claim the i just wanted to

say secretion in my definition because

it's gushy that was my my kids used to

say but um

auto catalytic and entropy secreting so

notice i didn't say it had to be

embodied in a specific form or specific

matrix or context it could be a market

it could be a markov you know a lot of

different things so um

so the reason i share that here it's

interesting because active inference we

have this definition of life like

corollas linnaeus you look at it you see
if it has feathers and you see if it has
blood and stuff like that but now we're
going towards mathematical definitions
of living things
and that's going to be very interesting
because i think what when we have a
working definition that works for
biologic forms so again my working
definition right now is auto catalytic
and entropy secreting
um then i can start to say okay
if that works for biologic forms let's
say that did work and maybe there's
problems with whatever but if it works
then you can start saying okay what
other systems happen to be auto
catalytic and entropy secreting and
therefore we shall now call living
and maybe we can start to look at them
and make observations about those
systems and what i'm thinking about i'm
a guy who's a lawyer and in finance for
years so i'm looking at this and saying
okay you look at these systems
biological systems came up with these
markets and these other
systems that we have to do things in the
world maybe these systems are
describable with biological framings and
so that's why the kind of work you're
doing feels like it as it starts to
reveal some more general definitions of
living forms taking those definitions
and then reapplying them back out into
the world and saying okay
are there living forms that have escaped
our detection as living forms and

basically are they is there scale
independence are there fractal elements
what are the things that we can now
describe in those things that we
previously had not thought of as biology
but in fact are describable as biology
that's very exciting to me so i just
wanted to say that the kind of things
you were describing there and that kind
of working way back down into a model of
life based on time irreversibility it's
very interesting i think thanks and very
interesting paper overall i'm looking
forward to

diving into this and the other that
other paper you were mentioning also
thank you cool yeah

yeah thanks a lot that that's really
great

so to tie this

from the some of the physical examples
we've been talking about like the
diffusion processes

to the

physics of information and information
flows

the big thing we've been talking about
is this statistical

physics and this bayesian mechanics
and we heard a little bit earlier how we
could talk about the moments like the

first moment the simplest form the a
posteriori estimation was the mean and
then the second moment of a statistical
distribution is the variance and there's

also higher statistical moments that
describe higher and higher aspects of an
informational distribution

there's also the concept in physics of moments like the position and the velocity in higher and higher physical moments

and so the mapping is that potentially these physical moments that would describe components of a physical flow that could also be decomposed using the helmholtz decomposition might also apply to some of these

informational

flows and decompositions so we could use those analogies like the the coffee mixing and the steady state of the coffee and then think about how a statistical distribution is playing by similar rules

and so it's also interesting scott about instead of going from the um non-living systems and then looking for that pattern and pattern matching to biological systems we can kind of start up on the top left with or something even more biologically fleshed out and then look for those patterns and pattern match

outside of systems that have you know a vertebra and a central nervous system because that shouldn't be the um the card to get into the club of information processing systems or entropy secreting systems so any thoughts lance

[Music]

yeah i mean

yeah i think i think that's that's really interesting um

i mean the the the real challenge here

is to make the link between the
physics and the information
and
the more i've been working on this
the more i realized that actually a lot
of a lot of physics or at least a lot of
statistical physics you can
um you can explain with properties of
information
so for example
entropy increasing is also
information being lost
um
not not
stochasticity
stochasticity so the inability to
predict the future exactly and
it is exactly the same as
entropy being lost over time
information being lost over time and
entropy entropy increasing
so i think there's um
there's very deep connections between um
yeah information physics
uh inference
and uh so some of them have been fleshed
out but not
i mean it hasn't it hasn't really been
put together at least as far as i know
in this context in the context of markov
blankets
in the context of how how can we
like have a theory of how adaptive
systems work
and
can we describe adaptive systems as
performing imprints
um so so i think this is this is really

exciting

and and also as you can see from this paper so the

the dynamics that we're considering

there they're extremely general

and for these dynamics for all of them

were able to derive some kind of

inference so this is to say that this uh

inferential description

is it applies to many systems

and so it means that on the one hand

it's very ubiquitous but then on the

other hand it's not it's not that useful

because

um

maybe like uh the things that didn't

look at all like biological systems

can still can still satisfy this kind of

uh free energy principle or bayesian

mechanics

so so i think what's really exciting is

to

to be able to

um in in the future to to like

specialize to those dynamics that look

more like biological systems and then

really look at these informational

properties

and i think a big one is

how much exploration do we have

and exploration is really something

exploration and curiosity it's really

something that characterizes uh

intelligent systems as we know it

and um

and yeah so there's some discussions in

in the literature of how we would be

able to use things like uh

information length and concepts for
information theory and information
geometry to quantify things like memory
and itinerancy curiosity and so on
so one question from the live chat
before you continue the question is what
is the explicit definition of
information flow and also maybe in that
if you want to describe a little bit
about this other
paper like what is information flow
what is information geometry and how
does it relate to what we're talking
about here
right um so so this
this is all the idea of um
as as an organism you're always encoding
some beliefs about your environment so
we talked about maximum posteriori and
variational inference
so the idea is that we're always
receiving sensations and then making
some inferences about the outside
now if you're making some inferences
about the outside
implicitly or mathematically you're
parameterizing a distribution
about things in the in the environment
and so information geometry uh then
comes into play uh information geometry
quantifies the extent to which beliefs
about the environment are different from
each other
um
so it doesn't it doesn't make sense so
for example
let's say that um
let's say that um

i think uh daniel's um
yeah uh daniel's uh eyes are blue
let's let's just say
um
and then i have some uncertainty around
the real color
um so that would be like one
one kind of uh belief i could hold and
then another type of belief would be
daniel's eyes
um
are brown and then i would have some
other kind of uncertainty
now the way um
now a question you could ask me is how
different are these two beliefs
um and if you have a if i had a
different belief about the color of his
eyes
would that believe be more different
than the beliefs i initially had
um so then information
geometry enables you to answer these
questions it enables you to say how how
two beliefs are different from each
other
uh by quantifying the distance between
them
but also even more interestingly
as i
um let's say i i've never met daniel
before let's say i haven't seen his
picture before now you give me a picture
of daniel
um i'm gonna make an inference about the
color of his eyes so from being
initially completely agnostic
i'm going to move my completely agnostic

belief to a belief that says oh maybe he
has
brown eyes
so this is to say that as we make
inferences and sample the world our
beliefs evolve and so information
geometry enables you to say um how it
enables you to quantify how much your
beliefs evolved evolved and what kind of
trajectory did they take
because i i mean i'm assuming at least
our beliefs uh they they evolve in a
continuous manner
as we sample more information
and then it becomes interesting to see
uh what kind of trajectories did my
beliefs take and was that was that
trajectory efficient uh if it wasn't
efficient maybe um i lost some energy
there that i could have used somewhere
else
so so
my my inference wasn't very optimal from
that perspective
and then so there's a paper about this
actually uh on active inference that
uses information geometry to to see how
how um whether inferences in active
inference are efficient from a metabolic
point of view so something that you can
look at with information geometry
thanks just to follow up from the chat
and then we'll return to our colleagues
here do we have
um or more precisely do we hope to have
a definition like an electromagnetic
flux in field theory
for information flows information flux

yes um

yes so so information flux is the
the movement of beliefs

and so you can

you can write that down mathematically

um

so so yeah i mean

a belief is a probability distribution

about things that are going on in the
environment

and as you sample information you would
change probability distribution so it'd
be like a motion

on the space of probability

distributions that's really the the

meaning of information flow but maybe

the question was about um can we derive

some kind of like maxwell's equation

some kind of law

about how um

about how information flow happens like

maybe an equation for information flow

and i would say that that's what we

that that would be the aim uh that would

be what what people are working towards

in theoretical neuroscience like some

kind of equation

that explains how we make inference

awesome

so a few raised hands scott first then

blue then stephen so go ahead scott a

couple of things that was fascinating

little bits there um so on that last

piece on the information for those two

points um the embodiment i'm thinking of

embodiment of information so i'm

thinking of a farmer that's deciding

what crops to plant so they look and

they get information from the newspaper
and they look at weather reports and
they gather information then they decide
on what seeds to plant the seed itself
is an embodiment of information they put
the seed in the ground they get results
with their farming the markets then
respond so there was there were seeds as
information farmers processing
information markets processing
information and one of the things that's
kind of interesting here it's a lot of
the work i'm doing now is in food chains
and trying to figure out how to de-risk
food chains and and just food generally
and
that efficiency question you just asked
is really interesting what are the
losses and conversions that happen when
information is embodied in those
different aspects of that decision
process through which something gets
converted from a need of society to have
a certain amount of corn at a certain
time of year
to through all those chains of seeds and
selections and farmers and processing
all those things it's kind of
interesting to look at the system as
both each piece of it is living the seed
is arguably living although there's a
question of where the seeds are living
among
people who are biologists but anyway
and then uh because they're not actively
doing stuff but the they're kind of
stasis but um anyway that embodiment
question is that's one and let me ask

i'll ask both and then turn off my voice
here so the embodiment question is one
kind of that notion you just talked
about about an overall formula maybe
there's an overall embodiment
characterization that's made possible
using those landscapes and looking at
kind of a multi-faceted landscape and
looking at those conversions that happen
from one embodiment of information to
another and efficiencies there that's an
interesting kind of possible thought the
other one is you said something about
curiosity before maximum exploration
maximum curiosity
that really struck me because one of the
things we've been advocating for i do
did a lot of cyber work a number of
years ago and people were coming out of
the military and we were saying oh hey
you ex-military guy you got to learn
more critical thinking and that was kind
of insulting because they were senior
people right they're like hey we can
work pretty quickly but really they
didn't and one of the things i started
saying is don't tell people they have to
have critical thinking that's not
consulting how they have to be more
curious
and it's kind of interesting that we're
getting now to these systems having a
maximum
exploration possibility a maximum
efficiency in some ways with curiosity
to me that really informs a lot of
teaching because if you have people who
develop a curiosity over their lifetime

they take care of being educated they continuously are learning and so that notion of exploration and curiosity seems like it could fundamentally through active inference affect how we do teaching and much less of rote teaching and much more of a curiosity based teaching because of the power you'll never know what you're going to encounter out there in the world and so telling your children hey be curious will intrinsically serve them versus teaching them a rote series of things to respond to their environment so just a couple of comments on those last two things i thought were fascinating thank you yeah that that's really interesting um so as you said and i think we all agree i mean information um or maybe 70 70 years ago or 80 years ago we didn't have any kind of information theory information theory wasn't a thing computers weren't the thing and we didn't really think about information even though it was there in books i mean i mean everywhere around us and i think with the advent of computers and information theory um just information became a big thing and people realized that it was important and now everyone thinks about everything in terms of information and and i think that's the

the right way to
to look at things i mean it seems to be
um yeah information can encode just
about anything by
by definition
um and so physics uh or at least our
descriptions of the world they're
they um
matter was the most important thing
matter and and waves uh this is how
it all started
and then um
and then information geometry and
information theory came into play and
people started looking at these things
in terms of uh information as well
but i think mata uh remained the big
thing
um
but more and more people are looking at
at information in terms of like the
the most important concept at least the
the root concept from which uh from
which you can like encode or explain
everything else
and um
[Music]
so so yeah information geometry came
around and it enables you to to see how
how information is flowing over time and
how that relates to heat and
things like that
and now
people are looking at how information
may be encoded by things about other
things
and this is really basic mechanics based
in mechanics the the root of the word is

beliefs about things that are held
uh that are held by other things
and i think this is um
yeah i mean basic mechanics
i guess started out i mean in a way it
started out long ago with like basic
imprints
um
but but uh bayesian inference started
out as a as an abstract uh statistical
concept and now people are trying to
relate that to to physics
um
and yeah so so i think we're moving
towards the physics that's information
based and
uh and based about uh yeah based about
beliefs about things
encoded about other things so this is
really basic mechanics information
geometry
um
and then what you said about curiosity
was also really interesting
and
so so yeah definitely i mean in ai uh
people are tuning curiosity so
um yeah they're trying to optimize
curiosity of agents with respect to
different environments
in uh in a paper that we have
called active inference on discrete state
space is a synthesis
we uh in in one of the appendices we we
justify the expected free energy
objective that people use for uh to
explain decision making in
biological agents

and we actually show that
we could have so the expected free
energy for for those who don't know
is basically ways exploration and
exploitation so exploration is really
the curiosity
so it has these two terms
and in general i mean these two terms
are put on the same footing
but actually in the in the appendix of
that
active inferences synthesis paper we
show that you could um weigh them in a
different way
and and it would
and i mean the
yeah the agent would still work somehow
but it will have a different behavior so
you could be a lot more curious or a lot
less curious
and that would be really interesting to
quantify in terms of like uh concepts of
information geometry
but so so this is to say that that you
can actually tune that and have agents
that have different curiosities
and also make me think about
what you said about things from
computational psychiatry
where um
people explain
diseases
like computational diseases as um
as people having like different priors
in their genetic model
um
and seeing how that how that would
affect behavior so you can

explain like sub-optimal optimal and sub-optimal behavior with like the same principles just by changing the model and we also have um a recent paper on the bayesian brains and the rainy diversions where but we're basically showing that by changing doing variational inference a bit differently you can actually um account for for agents that are optimal that make like the right decisions and agents that maybe are a bit sub-optimal in terms of curiosity and and so on so i think there's a like a lot of exciting computational work to be done in terms of characterizing different behaviors and i think we're really at the beginning and ultimately all these things will be able to i mean people will be able to account for the information geometry i think that's the right framework thank you lance just quick follow-up scott go for it yep quick follow two things when the beijing mechanics when you say beliefs about things that are held by other things that feels nice with my example of the farmer and the seeds because it's an embodiment each thing that embodies something is the embodiment is a belief in a sense right because it's holding a seed doesn't have a belief itself but it's embodying

information

and so

it's interesting to think about belief

of things and then the embodiment of

information in the thing intrinsic

affordance that the thing offers right

that is that what's the relationship

between belief and intrinsic physical

affordance that's in in a system that is

the result of an earlier information

system that's one follow-up then the

second follow-up is the way i soften

people up when you talk to older people

and you're like oh it's all information

it's not physical we're information

beings the way i soften them up just to

share is i say to them do you have a

401k or retirement plan most older

people are starting to think about that

and then they they say yeah and i say do

you think it's a pile of groceries

waiting for you to retire because it's

not it's just data in a system and if

that system goes away you got nothing so

it's that idea of getting people to

understand how much we depend upon

information and then people start

thinking oh wow i guess i really do

depend upon all sorts of other systems

in my life it's a something that

as physical beings it's hard for us to

realize how much we are information

beings and that's again just part of

that rhetorical exercise to get people

in a head space of saying wow maybe

things really are information that i

thought were solid like my retirement

plans but really they're just totally

dependent on systems and the information
output of those systems thanks nice
scott thank you
so um blue with a question go for it
so you mentioned a paper um that you
well i don't know if it was your paper
but about information geometry and
activated friends is it this one that's
markup blankets information geometry and
stochastic thermodynamics is that the
paper you're talking about uh no that
that's not this one so here uh if you're
referring to the one um
yeah about how um inference is efficient
like quantifying efficiency of inference
and how much like energy
um you need to do a kind of inference um
so that's not this paper
it's another paper called uh neural
dynamics
under active
inference um something like that
so i um have some questions and i don't
know anything about information geometry
i mean just a little tiny bit um so it's
something that i'm definitely starting
to get very interested in because i
think that um
it's gonna
be maybe like the next level of where
active inference goes maybe maybe i mean
i just have this suspicion so when um
when you have information and in an
information geometry like in an
information flow
uh i've been reading other papers about
information flow in channels right so
like when you and i mean it makes sense

because when you have when you know
something about the current state i mean
it's just like an inference like you
your belief about the next state of the
world is based on
the state of the world right now right
so

i mean it doesn't make sense that you
would think you know all of a sudden i'm
going to be like on pluto
based on my current information and so
it does is information geometry or do
the channels appear when you look at
inference in information geometry like
does it appear to be like a trajectory
um

that's a good that's a good question
so i think i think information geometry
is is more general like you don't need
to have a channel it's it's really about
um

at the heart of it it's really about
having different probability
distributions and these are actually
beliefs about stuff
and looking how they differ from each
other and like movement

uh how how um
probability distributions change over
time these kind of problems
um but then um so so you're right to say
that information theory started out with
the study of channels
communication channels between different
uh different things
um and you have a
yeah there's a lot of work uh on
information theory and channels and

still today with uh in engineering

um so

i would i would guess there's a lot of
work also on information geometry with
channels

but that would be like um a subset of
the whole field of information geometry
and i would say it's uh it would be
yeah i would say the people contributing
to that literature would be mostly from
the um from engineering or around
engineering

thanks blue anything there yeah go for
it yeah if i could just follow really
quick so the um

so i know like about information theory
and like the flow of information in the
channel say like a telephone line or you
know whatever like

some direct method of communication but
like the way i i don't know i don't know
if i'm thinking about it in the wrong
way because when i try to search for
like information flow in a channel

um and maybe it's the same but i think
about it in a different way like i think
about the channel flowing from
like the present into the future

right so like there's only so many like
next states that are possible right like
based on my present state and so is it
is it incorrect or is it correct to

think about like not a direct channel as
in i'm speaking to you over this

internet information channel but like
the channel from now into some future
state is that the same kind of like
mathematical engineering information

channel

right um yeah that um yeah you're

catching me off guard because i don't i

don't know about channels

um so i mean what you say sounds sounds

possible and i would say yes

but i wouldn't be able to say for sure

just because i don't know that

literature

thanks well

i'm sure come back to this in the

weeks and years to come so stephen

go for it

yeah i'm

i'm really interested by the um

these different um

tying these bits together in a way i

suppose that sort of come up um we've

got

this sort of

information geometry that i suppose can

come from an ergodic space or a kind of

a state space

and then you've got solenoidal flows

within maybe an iso contour of that or

potentially i suppose it doesn't have to

be solenoid or it could be like an iso

contour

flow if it was more complex

and

when you've got that potential

to um

go and have a solenoidal type flow to

sample

would that be

um more a slightly more of a second

order

sampling approach that something might

use so you've got you know the
distribution could just emerge from a
steady space steady state sort of
ergodicity process and you would maybe
somehow

systematize a flow or way of engaging
that information and i could i could
imagine that would then

make sense of some of the entropy
entropy consuming potential of a living
system which is that it can kind of
go upgrade in descent and
the solenoidal flow then
gives scott's

sort of entropy secreting option it
seems a bit more viable then because
maybe i could choose
multiple solenoidal samplings
of

information geometries and
secrete the ones which aren't as good if
that makes sense so um so i'll be
curious in terms of how

you know that this the ability to go to
more second order
understandings within a system
um and within the dynamics might be
offered by

um
the solenoid or sampling maybe even
taking into account things like the the
landscape used on the um
the mountain car

um
problem you know that landscape if you
can't sample the whole landscape
ergodically could you have lots of
solenoidal

samples or something like that around
the landscape

yeah um

yeah it's a really good point so there's
deep connections between so you know
flow and information geometry
and mostly in the context of the as you
pointed out the sampling literature
so in um

so the the idea is that if you start a
stochastic process somewhere
and you have solenoidal flow
it will converge to its steady state a
lot faster than if it didn't have
solenoidal flow

so then you can quantify that in terms
of uh information geometry maybe taking
a shorter path to the steady state
um and things like that
um

so then there's also um a really nice
new paper actually called um
is there is there an analog of nesteroff
acceleration for mcmc
and so that that paper shows that
um

yes

that paper really like does a nice job
of illustrating some some deep
connections between
uh stochastic processes and information
geometry and so the idea is that if you
start with a very simple um called a
very simple process called overdamped
knowledge of dynamics
then you can show that um
actually the the dynamic
in information space

is doing um so like the process is doing
a gradient descent on kl divergence
um to reach to reach the the steady
state so the target distribution
so this is just this process is exactly
doing like standard standard variational
inference
but then if you add some solenoidal flow
you actually
make that convergence happen faster so
it's like an accelerated method
and so and so in terms of biology you
can say that if you have solenoidal flow
um your inference happens faster so
you're like uh
you're able to infer things a lot faster
so you like in a sense more intelligent
i think we talk about it in in one of
the figures
in in the part of the paper
and so and so in that paper that i just
mentioned
uh they showed that with a particular
stochastic process
actually
implement some kind of
accelerated variational inference
um
so yeah there's some literature about
that and it would be interesting to see
uh those processes that perform
inference very fast
whether when we put them in the bayesian
mechanics context whether we can say oh
these actually
can be seen to correspond to
to biological systems that we've been
studying things like that

okay i just asked one quick question on that

when you say add in solenoidal flow is that adding it in the sampling methodology or is it somehow maybe i suppose it's a bit could it be thought of physically or is that how does that work

yeah definitely so the the sony nodal flow is really this time irreversible part

um so so

intuitively it's this kind of like stirring or mixing the system so maybe if you have um

a really nice example actually is if you have a cup of coffee

and you and you pour milk in it

now what you could do

is just wait for the milk and the coffee to mix together

and this would be no solenoidal flow because you're not mixing anything like

it's just happening on its own and you have a dynamic like in the in the figure that we saw before like on the bottom left so it would be time reversible

and this would like actually converge the steady state pretty slowly so in terms of information geometry you would have a slow dynamic to the steady state in terms of statistics you would have a slow variation of inference because you reach your target distribution very slowly

now what you can do is adding in solenoidal close so this would be i take my spoon and i'm going to

stir the coffee and the milk together
then we all know then
that this is going to allow
the system to mix
and to and to
actually become uniformly mixed a lot
faster
so so mathematically
it means that we reach the target
distribution of the steady state faster
so the information geometry we're also
faster in terms of statistics
you can interpret that as doing
variational inference faster
now this interesting thing about adding
in solenoidal flow
is that um
in physics it's called breaking detail
balance
and breaking detail balance is about um
you actually probably need to to input
from entered input energy from the
outside
um so so this is the physical meaning of
the time irreversible part of the
dynamic
when when there is sometimes
irreversible part it means that the
there is
actually energy coming in from the
outside
and and this is also like um it ties in
nicely with biological systems we know
that biological systems consume energy
from the outside
so we know that they have to be
that they need to have some kind of
this irreversible dynamic

thanks lance so

anyone else in the live chat who wants
to ask a question this would be a good
time to go for it and then dean
and then anyone else

let's be gentle on me what i'm about to
put out there is i just feel really
vulnerable but i'm gonna throw it out
there anyway so

lots of chat chatter about the encoded
aspect of this and i think we all kind
of understand what that means
so

i'm gonna i'm gonna raise a metaphor
kind of a

around negative space and the idea of
threading a needle

which is if you think about it as a just
as a distributive exercise is plenty
complex like there's a lot of math in
terms of being able to just do that
thing that's that's hard hard enough
so

if i'm trying to explain to somebody
else a bayesian manifold and what that
represents

can i say that it's both a barrier and a
filter at the same time can i say that
it acts kind of like a sieve
meaning

meaning that it's mathematical structure
kind of acts as a ground to the physical
things that we

are inferring are going on in the world
so i guess what i'm asking is is
statistical mechanics

serving
to

allow us to understand what negative
space is as a non-form
within all
as as part of all the forms that we're
seeing the edges of and
and having a liminal understanding until
we get to the very limits of what and
the high definition of what that means
so
does the statistical manifold allow us
to go in around and pass through zero it
doesn't allow us to reverse the spin of
the solenoidal
but it does allow us to get away from
zero and move closer to what the known
is
because
if i if we continue to kind of go back
and forth between the physical math and
the statistical math i know why we have
to kind of talk
both at once like we have to have two
hands to clap but i think it's i think
what ends up happening is
in this conversation the encoding and
the physical piece tends to dominate
and then and the non-physical stuff the
statistical stuff maybe that ground that
the statistical manifold
allows for it doesn't it hasn't maybe
been fully fleshed out
in a way that helps me
help explain it to other people who
aren't as far along in this journey as
you are and i'm hoping to get to a place
where i can
translate it for them so maybe you can
just talk to that a little bit about

what is the statistical manifold is it
just the hard stuff that we see or
should we be paying attention
to maybe the spaces between all of those
data points
right um
yeah so
that's a a really interesting and
actually a hard question to answer um
but just to start with the the easy part
the the statistical manifold
is the mathematical object
that that says these are all the beliefs
all the different beliefs that you could
hold about your environment right and so
as time
as time evolves and you're gathering new
information
your
what your brain is doing is moving on
this statistical manifold your beliefs
are changing
over time and then in terms of um
explaining it to people
so the
always the the way the way that i try to
explain it is that
this um this like in information talk um
things encoding
things
things encoding information about other
things
it's just a different description
of the world
so we have a description that that takes
like matter and waves and forces and
things like that and then we have a
different description that looks at

information

and i think these two descriptions

are equivalent

um it's just two different ways to talk

about the same thing

on the other hand this um

way of talking about the world in terms

of information flows and information

processing

i think

it might be more general in the sense

that instead of needing to have like

matter and

all these different concepts different

kinds of particles and it gets really

messy

maybe it would be possible to unify more

with the concept of information because

with information you can explain what a

photon is and what

what a proton is and and you name it so

so it would be more um

more unified and maybe more

mathematically simple at the end of the

day but that remains to be seen

um interestingly so so there's this guy

evolved from from uh the computational

engine evolved from alpha so

this guy this guy's actually a physicist

i didn't know that until very recently

but he

has been doing some really interesting

work over the past few years

working towards a theory of everything

that is based on the concept of

information so information and

computation is the

like the the thing he starts with

and i think he's managed to
um he basically postulates that the
that the um
the world
evolves in time
according to some kind of computation
that's recursively applied to itself
so
you you could think of it in terms of a
dynamical system but so it's a like the
way he thinks about it is a bit more
general but it would be like maybe
um you give a number to a program and
then the program outputs a number and
then so that would be one time step
and then you reapply the program to that
number that would be another time step
and so he can he
and so he postulates that the universe
works in that way
um it evolves um according to like
discrete steps
with a repeated program and the program
would be the laws of physics it's like
kind of an abstract informational way to
look at it
and so the
so yeah so he he makes this um this very
abstract approach to physics
and uh and and also very simple i mean
very kind of unified approach to physics
and i think he's managed to derive
approximations to the equations of
relativity
just with that so with a certain kind of
program i mean anyway he's been um
doing some really interesting
work trying to unify known physics with

the concept of information
and he's written a very nice uh
article like a blog post about it i
think if you
if you google like a wolfram theory of
everything you might you might come up
it might come up
um but yeah it's um
i mean it's not it's not technical i
mean it is technical somehow but it's
not like
there's no equations it's conceptually i
think conceptually hard what he does but
the way he explains it is nice
um
so yeah i think we're moving to this
kind of
information-based physics but doesn't
mean that the physics that we know with
matter and so on should be discarded
uh it's just like different different
levels to talk about it just like we
have
um psychology that talks about the brain
uh from a
like very overall perspective
and then we have like
yeah talking about like maybe biases uh
our biases decision making so that would
be one description of the world and then
we have
molecular dynamics that will look at the
brain from a very
um
from a basic molecular level
um
and and all these descriptions have to
fit together somehow

and i think depending depending on what
we're trying to ask uh
one description would be more fit than
the other so
if i'm studying a human psychology is
probably much more useful than molecular
dynamics
thanks lance one quick comment on the
information is it allows us to drop into
a mode of matter and organization as
information
um which is still within the objective
stance but also it allows us to model
ourselves and our beliefs and thinking
about them informationally about other
beliefs or about matter which is a way
of mathematically
complementing the relational stance
so it also moves us towards wolfram's
vision of what new science would look
like because we actually have
quantitative relational techniques that
help us deal with our uncertainty and
about the way that we interact with the
world actively
interfering and measuring and
manipulating
not just like us walled off as the
investigator trying to get to the most
exact
framing of a system so it's really
interesting developments so we have time
for some more questions so scott
and then after that we'll go to steven
so go ahead scott
a couple of little thoughts on that last
bit so that that
multiple dynamics that you were talking

about very interesting it got me
thinking about
that coast of england problem
from um fractals where if you measure
the coast of england with a one mile
yardstick versus a one inch yardstick if
it's a one inch yardstick the coast of
england is longer
right it's because you're measuring more
than nooks and crannies right in the
fractal uh coast so
that scale
dependence when you talk about the
dependency we see
nature and see biology through different
lenses
it's kind of interesting if we have a
unifying lens maybe you start to have a
fractal and scale independent
descriptors then we'll pop out of that
because you'll start to have things
that'll
anyway it feels like this skewer coming
at descriptions of systems as you were
saying from different in different ways
they're the same system ultimately and
so there'll be an interesting analysis
on the way in which they come together i
guess is that this one piece another
thing
the
mention about information and the um
that kind of wolfram idea that expansion
um one of the things i had a
conversation with a couple of um
astrophysicists and i was talking to
them about
um seth lloyd out of mit a part of his

book called programming the universe he
talks about
the idea that all interactions since the
big bang have increased exponentially
just the number the volume of
interactions
and
i was talking to somebody about it and
they said well
if the universe is made of information
then the exponential increase in
information maybe that's what dark
energy is we perceive dark energy that
expansion force as being
something but really what it is is the
universe expanding out exponentially
because it's made of information and so
something just to think about it kind of
leaps around there between physics and
information theory but there's um
that's one other piece of it and then
the other piece of it was this
idea the outside energy you were talking
about before the need for outside energy
one of the things that i've been fussing
around with is whether in fact and in
2013 i did a presentation mit called
entropy accounting
and then the year after they did
something called entropy engines and the
idea that was that was that in
information theory in carnot's equation
you need a hot and cold differential to
perform work
um carnot's equation those differentials
are identical to arbitrage differentials
you need for a market to perform to move
if there's no information differentials

you don't have trading in markets and so
one of the questions is
roger penrose wrote a book recently
about the time before the big bang what
he talks about is we have the yellow
light of the sun come in and the
infrared light go out we don't overheat
except for climate change and carbon
dioxide blankets
because the net energy gain is zero
right energy and energy otherwise we'd
overheat he said what we gain
is neg entropy because the yellow light
of the sun has a higher information
carrying capacity than the red light
that goes out plants construct
themselves from the yellow light of the
sun
create neg entropy and then we
now
projecting forward are now burning that
fossil fuel that stored up that neg
entropy and releasing disorder
in the form of
now climate change disorder
political and social disorder so where
i'm jumping around there is suggesting
that disorder
accounting
and entropy accounting when you have von
neumann and shannon's link of physics
and information and other links like we
have here
that maybe when we're talking about
entropy as disorder we can talk about it
not by analogy but direct application to
social contexts
and start to use information and

understand disorder in society and in
markets as uh amenable to the same kind
of math that we find in some of these
biological systems thanks just a few
thoughts
thanks scott lance if you want to add
anything
sure um yeah that was really interesting
and um yeah so
um i want to add actually that
uh all this um
the thing that we talk about about like
reality can be described at different
scales
and i think there's a nice extension to
this work that that speaks to that you
could
um if you look at a human i mean it's
composed of a lot of cells and each cell
has its own boundary and then these
cells form organs that themselves have
their own boundary so there's um
let's read this recursive aspect of like
markov blankets being assembled together
at different scales and i think um
i think it would be interesting
interesting to impact that because i
think that's how
biological systems are formed but also
societies are formed as collections of
multiple different individuals that are
themselves that themselves form
organizations that
um that talk to each other and form
together a society so it's really this
recursive aspect and
being able to hopefully use the same
kind of math to talk about all these

different aspects

and

and also i mean

people um it's true like a lot of people

in the past maybe 10 20 years maybe more

they've used the same kind of

dynamical models that we have in this

paper i mean so very very standard

models to

model for example interactions between

people

and in particular models of opinion

formation

so for example maybe let's say we're in

the us and some people are democrats

some of our republicans some cannot

really make their mind yet and so they

would talk to each other they would

interact

and then they would converge onto a

steady state

where some

yeah some some people would be die hard

democrats in that city state so they

would kind of stay in that area

some people would be die hard

republicans and some people would like

yeah fluctuate flock to it in the middle

um so

i i think we can use

i mean

this math about stochastic processes and

information is really the

um

yeah the the

ground i mean the

the

a good framework or useful framework to

to model many kinds of things

um

but then

i think what a lot of these models are

missing is really this belief based

aspect

so a lot of these models of opinion

formation there they just say oh i'm a

particle you're a particle and we

interact just like particles in physics

with interact

um so so it's quite simplistic um

so i think once you i mean uh things are

very small

maybe we can describe them in a very

simple way

um but as you as like marco blankets

self-organize at different scales and

actually

as you go to larger and larger scales

um

things become a lot more complex

and i would say maybe we are the scale

at which things are

most complex

or molecules tend to be quite simple we

tend to be quite complex and then if you

go up and up again

maybe to planets

then things start to become really

simple again

because all these random fluctuations

are kind of average average doubt and

then we have newtonian mechanics where

everything kind of looks deterministic

so i think there's this like

sweet spot we're in where we're not too

big and not too small

and interesting things happen there

um

and that's where i would say that's

where i would say bayesian mechanics um

is hoping to fit in

in terms of uh explaining and describing

what's going on

cool so stephen go for it

yeah

that that um that idea again that brings

us back to embodiment i suppose because

we're embodied at

a particular size and scale and

um they sort of say if you go right down

to the smallest scale

of like and then right up to the biggest

scale

we are kind of in the middle

um

ironically but so i was wondering how

sort of following on from what you're

saying and bringing in the computational

psychiatry piece that you sort of

alluded to to this is

if we if we say that we've got

everything that we perceive even the

physicality of the world

we actually only have access it through

information anyway even if we are

actually accessing the the true physical

world

so we've got this process of sense

making that we're doing to to make that

idea of what is our consensus reality so

if our kind of working space that we're

in is our consensus reality

what's interesting is that

with

psychology and social sciences

you can almost say psychology is a bit

like

the predictive processing

model of the world it's like these are

predicted ways that are out there

um that have been chosen as the models

within a particular sort of discrete

spaces

and those are externally managed so it's

like an externally managed

system entry percent there could be an

entry repeat secreting system where it

looks like there is although entropy is

not actually a substance to secrete per

se

and you could say that it's externally

managed information geometry that the

social science as well with the averages

and the psychometrics that they

reference to they're all sort of housed

out there in the world and you've kind

of got

the kind of what we see is our reality

in the phenomenology of our experience

is kind of the boundary but what

your work's doing and what computational

psychiatry does and which we're not able

to do without those tools

is it allows us to go into that

neurobiology

and the actual like the internal

internally managed

stochastical dynamics of our own models

so to speak so not the ones that we make

out there to be our perspective on

which we try and reduce free energy

around but just what is the models that

we're doing what you call entropy
consuming i we're actually
just swimming in that entropy and
finding the variations right
and so i'm curious like
that that that ability for this type of
work to
contribute
to a new kind of
um way of knowing about
what it is to be in the world and go
into our internal models rather than
be reliant on these external
psychological and social science models
what's your thoughts on
how this might link into developing that
that field

[Music]

so
so are you talking specifically about
the link between
all this physics and and the psychology
or or is your question a bit more
general um well maybe the the boundary
between the kind of
like the scales of basically the
boundary of going for externally making
models of the world and standard models
like psychologists are doing and social
sciences are doing and psychometrics and
the ability to actually go into our own
models our own phenomenological
neurobiological
models and just say okay what are the
dynamics
that are alive not what are and
how we how are we actually managing not
how do we manage our models of what we

think they are which is kind of what happens in our kind of niche but we're able to probe our own phenomenology so i was wondering how you see that tying into that yeah um yeah that's a that's a really good point so so as you as you point out i mean now models in like psychology and and uh computational psychiatry i mean they're driven by intuition they're driven by by experiment and they're driven also by your own biases um and our observations about um yeah patients who who suffer some some kind of um yeah some kind of let's say schizophrenia and so we know we know schizophrenic patrons behave like this and like that and so we try to create models that kind of reproduce that and then we try to um to validate these models by saying oh uh maybe i can simulate some kind of uh action potentials of some neurons and then i'm gonna check whether actually these organisms that i'm studying have this um same kind of action potential the same kind of electrophysiological responses um so i think that i mean i think that's really good um and and the research is doing a lot of

progress

um

but then

this kind of work not not this paper but

this line of work

would be able to to like

go the i mean um

i mean it takes a different perspective

and would be able ultimately hopefully

to describe mathematically

um what thinking like a human is uh what

what is being human mathematically and

of course there won't be like a

definition of this is a human

but they would hopefully be in the

future a mathematical description of how

we make decisions

and what class of models are we

implementing to make sense of the world

and how can we characterize these models

in terms of and and hopefully um the

community will get to that through

through physical experiments but also

through like

just mathematical conceptualization

um i think using using mathematics you

can

you cannot get you cannot just get there

without any like kind of experiment but

you still can

[Music]

there's still quite a few constraints um

that we have on

that we know that that humans have looks

for example breaking detail balance um

creation of entropy all these things we

we know

we know humans satisfy like uh

i mean they have a lot of um
characteristics and we can incorporate
that in the physics and i think this
really constrains the model space
that then people in psychology can go
and look into
so it just feels like um
when you're digging a tunnel and
psychology would like kind of dig in one
way
and basic mechanics would be digging
from the other direction and at some
point
uh they will meet in the middle
hopefully um but i don't i don't think
they have met really yet and it will
take a while
awesome
um
what
a fun discussion and it's so cool how
this paper it's almost like there was a
few stairs
we we explored
some basic ways to frame
how
nodes on a bayesian graph insulate other
sets of nodes
and then how that conditional
independence can be finessed for these
increasingly developed and elaborated
networks that do
as we talked about
expectations over external states all
the way up to variational base and then
it's like we at the top or at the middle
of that staircase started taking it in a
lot of other directions so this was

really a fun discussion
just in the last few minutes
maybe we can
any and all of us think like what are we
gonna
continue working on for some of us
that's probably a research project
others that's
i don't know lunch or something else
just how are we going to keep on
learning about this
and
start to apply it in our own work or
research
so i have a really good question liz are
you working on um
the
next paper where you're going to
partition the external state space or
have you started or it's just like
clicking on a backburner somewhere
uh it's not cooking yet so so you can
uh you're more than welcome to to go
ahead and
and start plowing
it's at the farmers markets the
ingredients are still out there and dave
also raised in an email some really
interesting questions about other
partitionings
and um
basically philosophical threads that
might be translated into different
possible partitionings
within internal states or
other ways of partitioning because
it was the partitioning of the blanket
states from kind of one set of

blanketing states into incoming sensory
and outgoing action active states that
allowed for this perception as inference
action planning as inference
step

and it's not that that's the end game of
node partitioning

so there's so many ways that these
partitions can continue stephen
yeah i'm gonna be

i'm trying to see how i like that
analogy the two tunnels or whichever way
the maybe it's a whole load of like um
insect warrens or whatever it is trying
to meet and um so psychology and
bringing together some of these
processes so i'm going to try and bring
that together maybe more actually not so
much from a clinical perspective but
maybe more from a community psychology
and psychosocial psychosocial
methodologies that might be used so
um

i i'm going to be chewing on that as
best i can so maybe if i ever stuck it
might i might fire you an email if
that's okay

yeah um

there's many ways we can continue and
also i think all along our trails we
dropped a few little notes for some
other pieces that could come into play
if anyone else wants to raise their
jitsie hand

we can take a last thought
otherwise though just in closing lance
this was awesome and conor as well for
joining the first live stream this was a

great

uh development we all learned a ton so

i hope those watching live or in replay

also

got something out of this paper

you're always welcome back anytime you

want to jump on a guest stream or be a

participant just

always welcome

thank you so much daniel and thank you

everyone for the discussion i really

enjoyed it

okay

see everyone next time

thanks bye

[Music]

great times thanks everyone